

Lecture -8

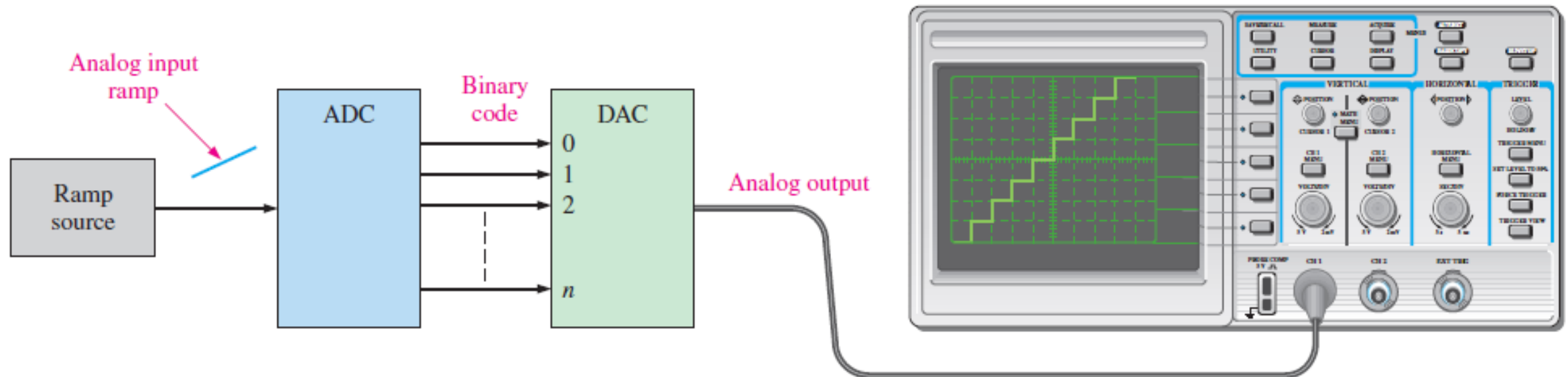
Introduction to Digital Signal Processing: 2

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Introduction

- Digital to analog conversion is an important part of digital signal processing.
- Once the digital signal has been processed, it is further needed to be converted back to the analog form.
- A Digital to Analog Converter (DAC) is an electronic device, often an integrated circuit, which converts a digital signal to its corresponding analog signal.
- Two types of DAC will be studied:
 - Binary Weighted DAC
 - R-2R Ladder DAC



Binary Weighted DAC

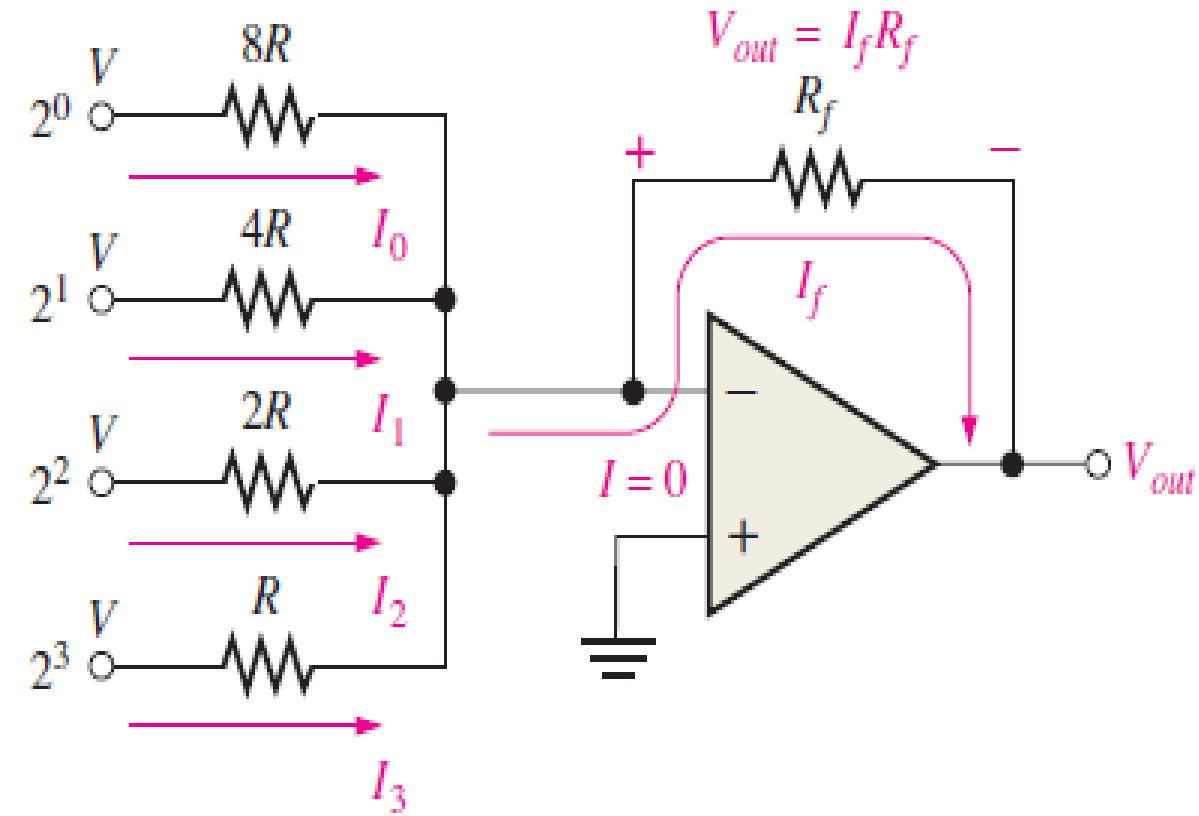
- The figure shows a 4-bit DAC.
- The resistors have a specific value depending on the binary weights.
- The resistor corresponding to MSB have the lowest value
- The resistor corresponding to LSB have the highest value.
- The output depends on the presence or absence of current in the branches.

$$I_0 = \frac{V}{8R}$$

$$I_1 = \frac{V}{4R}$$

$$I_2 = \frac{V}{2R}$$

$$I_3 = \frac{V}{R}$$



$$I_f = I_0 + I_1 + I_2 + I_3$$

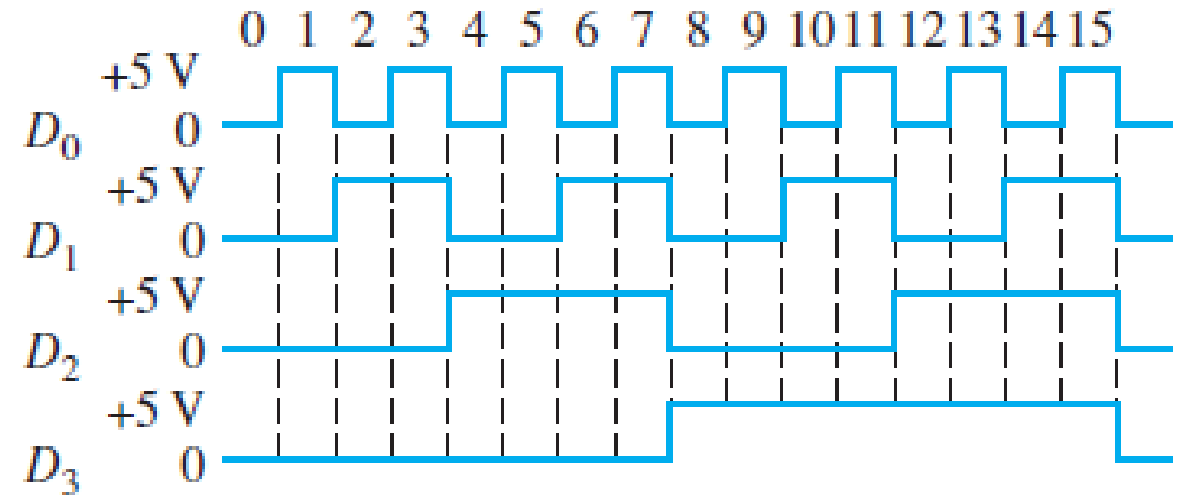
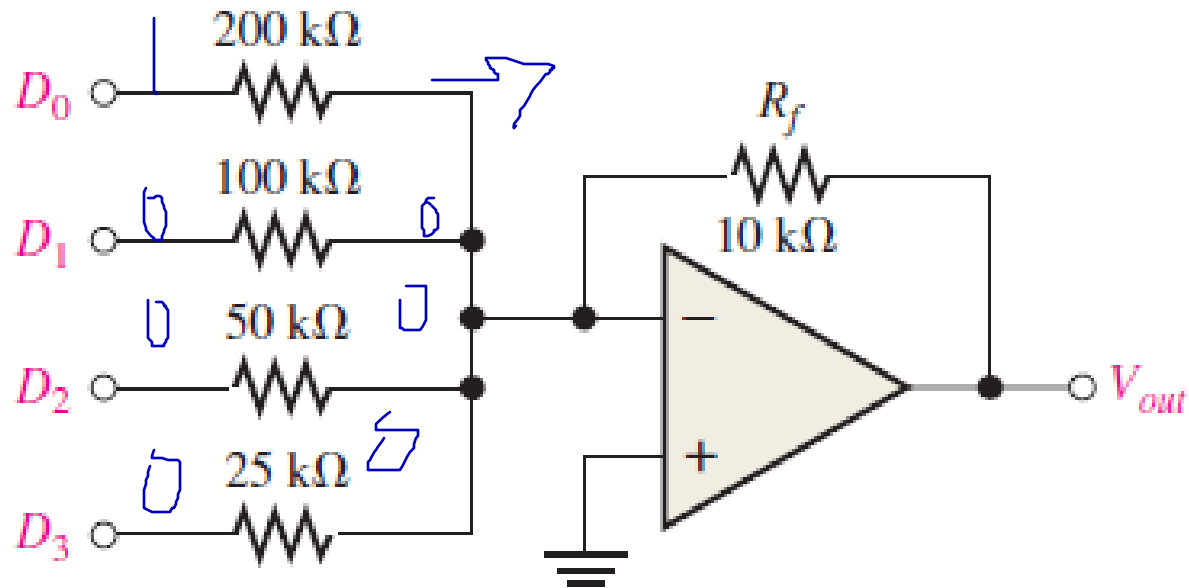
$$\frac{V_0}{R_f} = \frac{V}{8R} + \frac{V}{4R} + \frac{V}{2R} + \frac{V}{R}$$

Now if $R_f = R$

$$V_0 = \frac{V}{8} + \frac{V}{4} + \frac{V}{2} + \frac{V}{1}$$

Binary Weighted DAC

Determine the output of the DAC if the waveform representing a sequence of 4-bit numbers are applied to the inputs. Input D_0 is the least significant bit(LSB)



Binary Weighted DAC

- First we need to determine the current for each of the weighted inputs.

$$I_0 = \frac{5V}{200k} = 0.0025mA$$

$$I_1 = \frac{5V}{100k} = 0.05mA$$

$$I_2 = \frac{5V}{50K} = 0.1mA$$

$$I_3 = \frac{5V}{25K} = 0.2mA$$

- Now we will calculate the output contribution for each of the weighted inputs. Since no current goes through the inverting input, the current only flows through the feedback path.

0001

0010

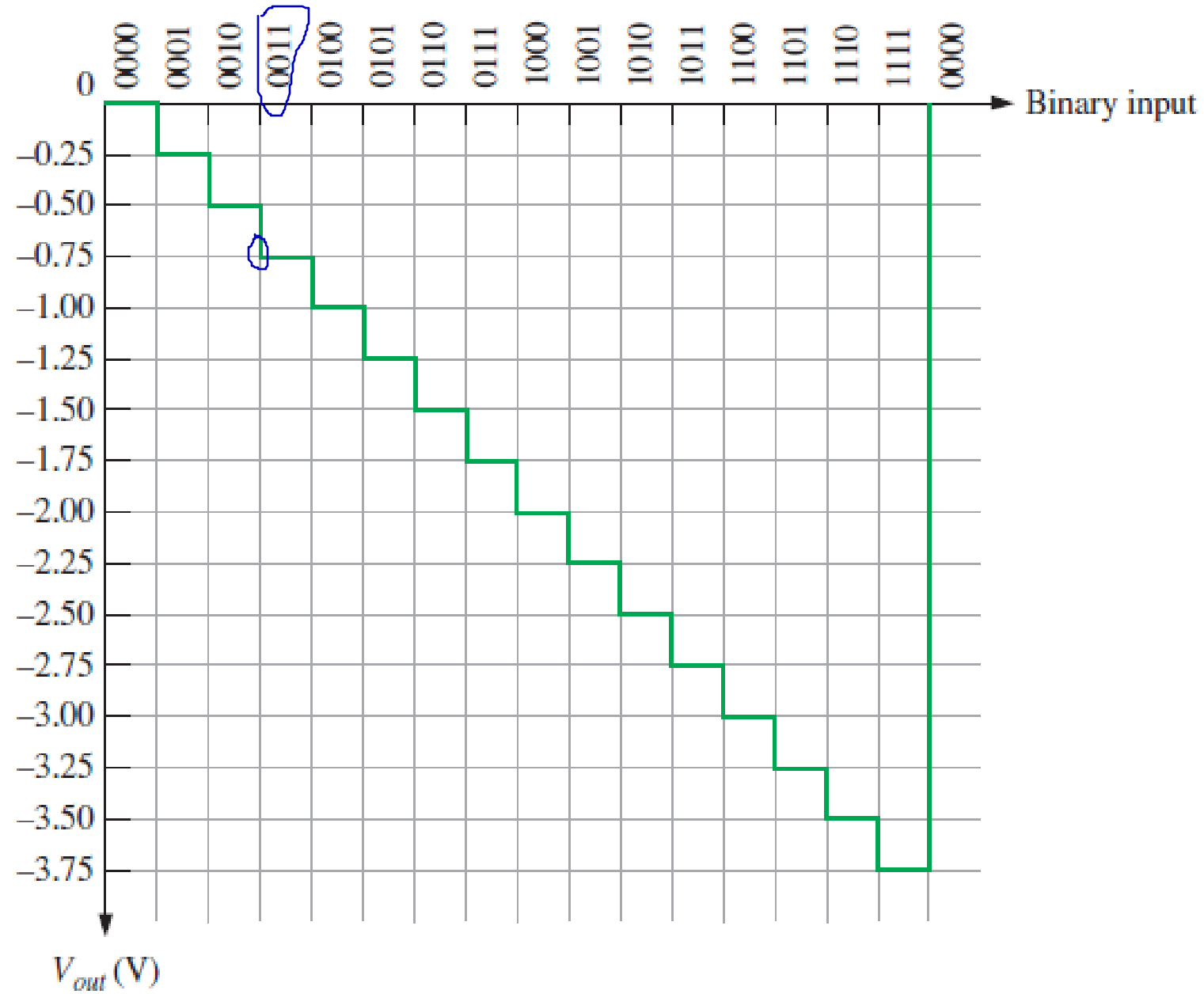
$$V_{O(D0)} = (10k)(\overset{R_f}{-0.025mA}) = \overset{I}{-0.25V}$$

$$V_{O(D1)} = (10k)(-0.05mA) = -0.5V$$

$$V_{O(D2)} = (10k)(-0.1mA) = -1V$$

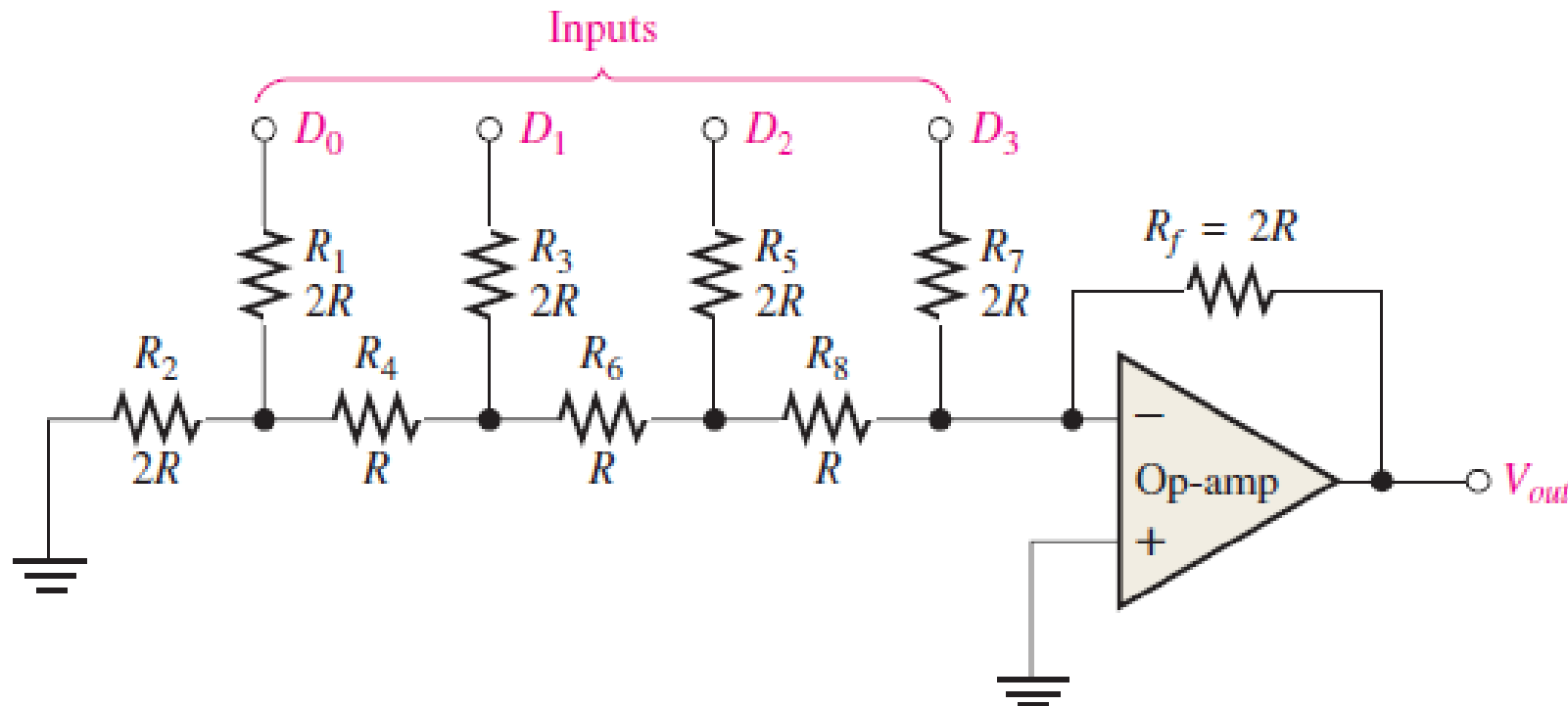
$$V_{O(D3)} = (10k)(-0.2mA) = -2V$$

Binary Weighted DAC



R-2R Ladder DAC

- Though binary-weighted DAC is simple to understand, construction of it is hectic.
- For binary-weighted DAC we need exact multiples of the resistors (R , $2R$, $4R$, $8R$, $16R$).
- For DAC of higher bits, the construction becomes practically infeasible.
- However, R-2R DAC solves this problem.
- The construction only require resistors of value R and $2R$

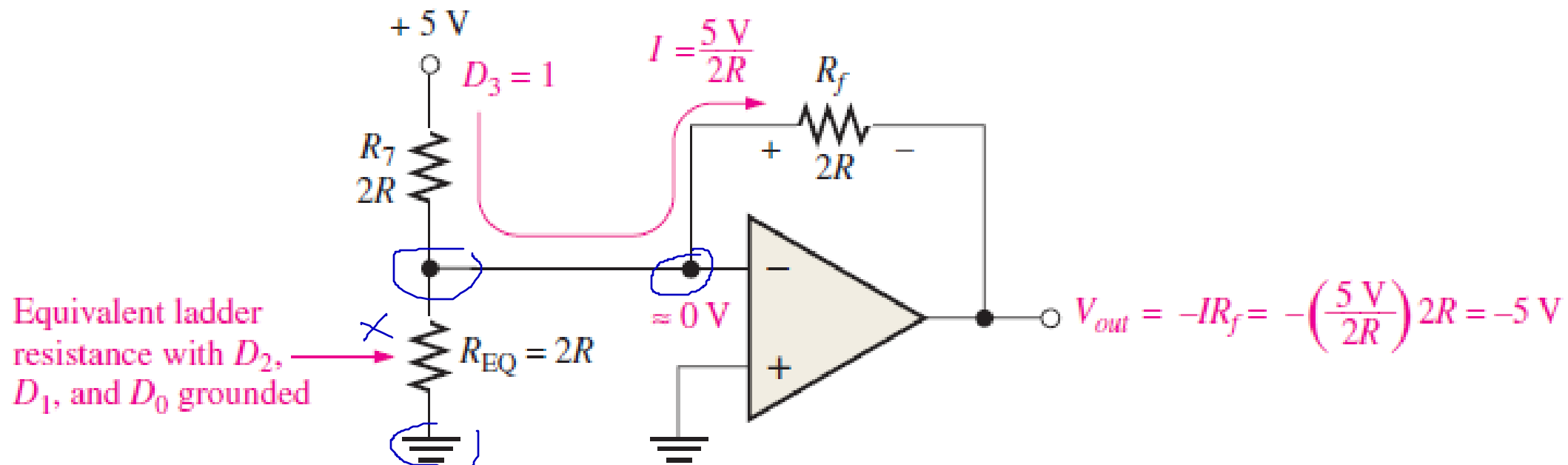
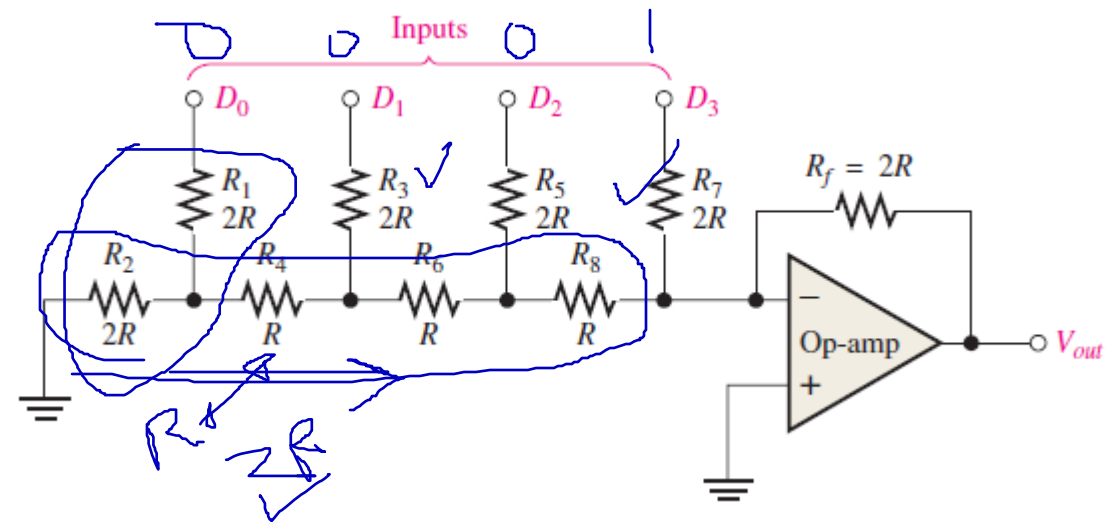


R-2R Ladder DAC



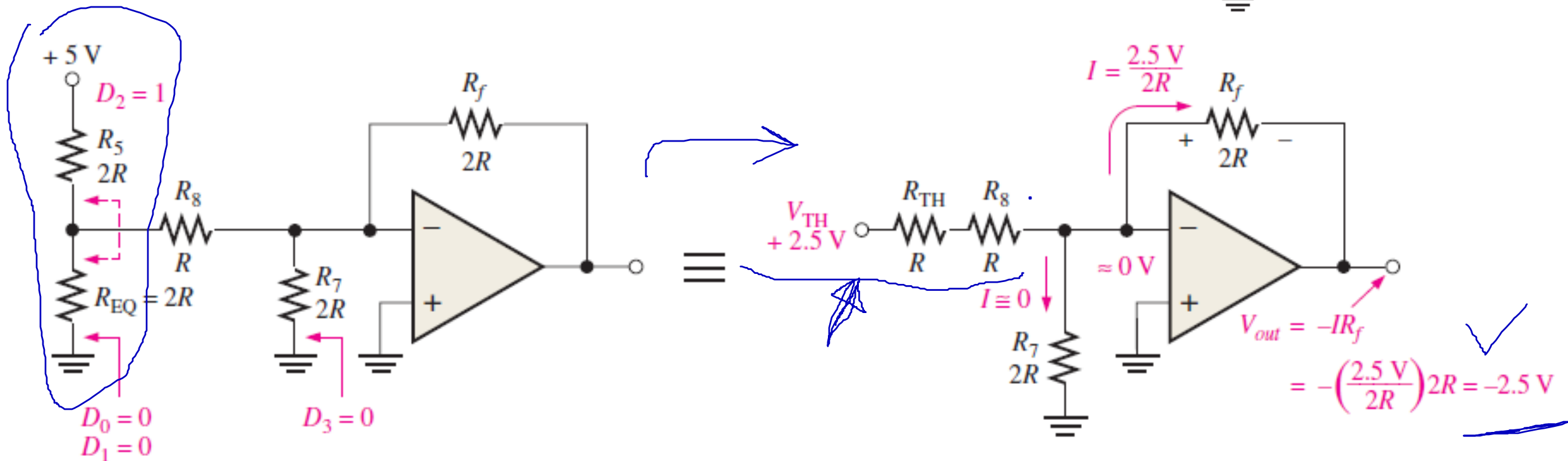
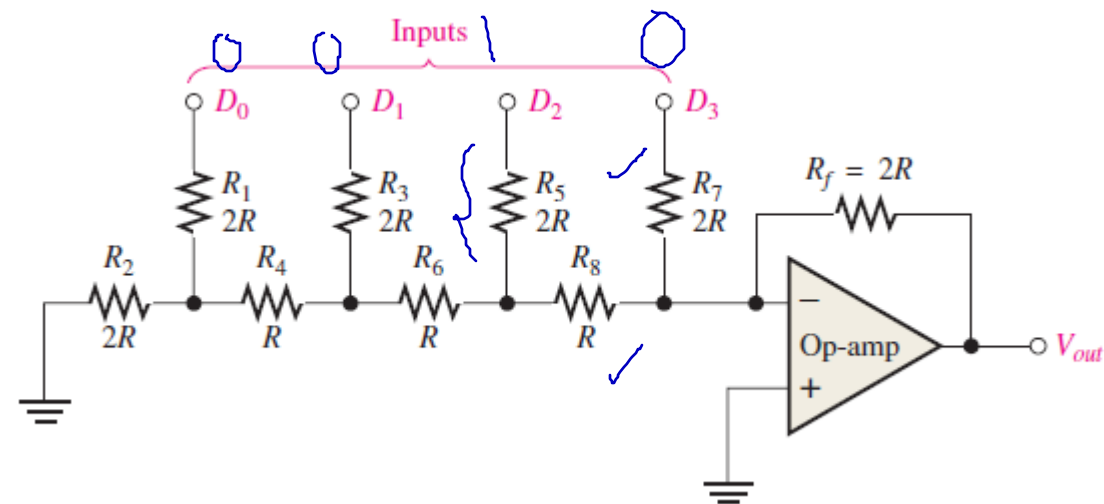
$1000 \rightarrow 5V$
 $0100 \rightarrow 2.5V$
 $0010 \rightarrow 1.25V$
 $0001 \rightarrow 0.625V$

Contribution of D_3 .



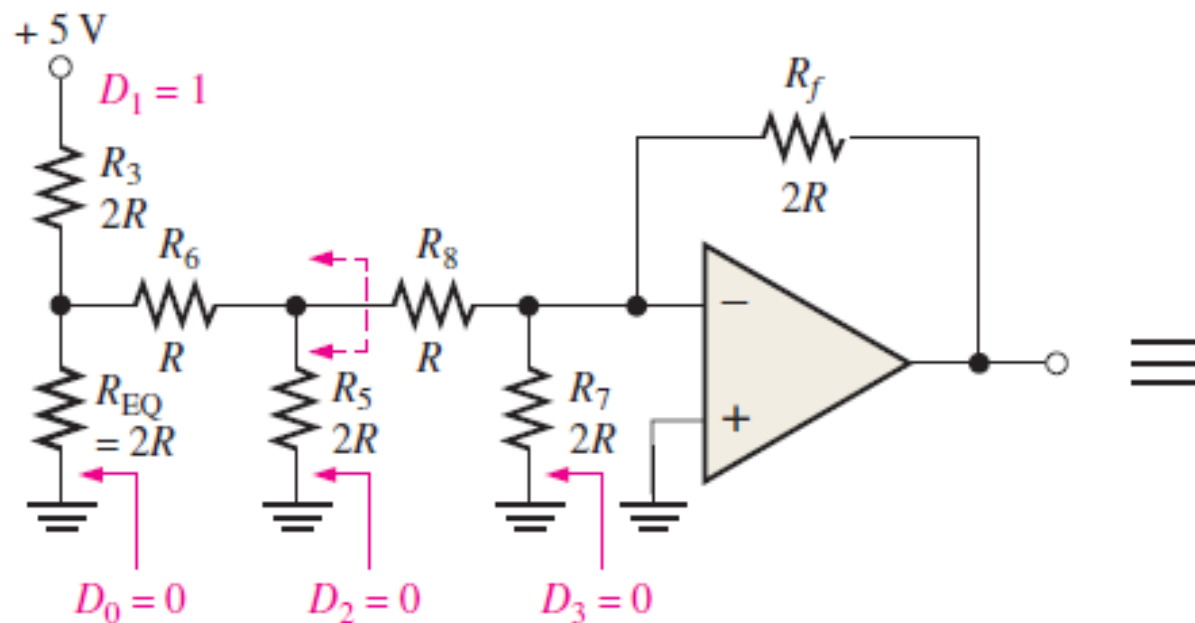
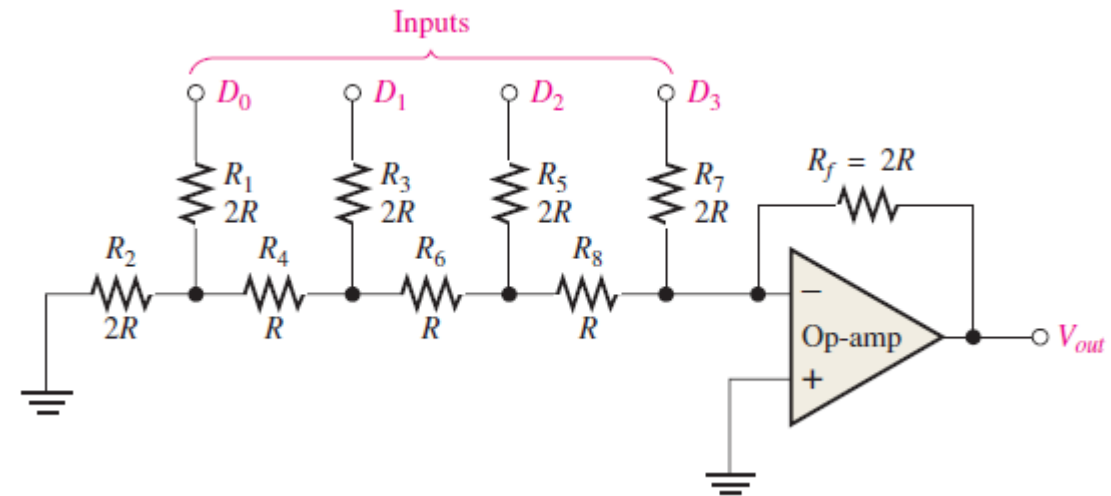
R-2R Ladder DAC

Contribution of D_2 .

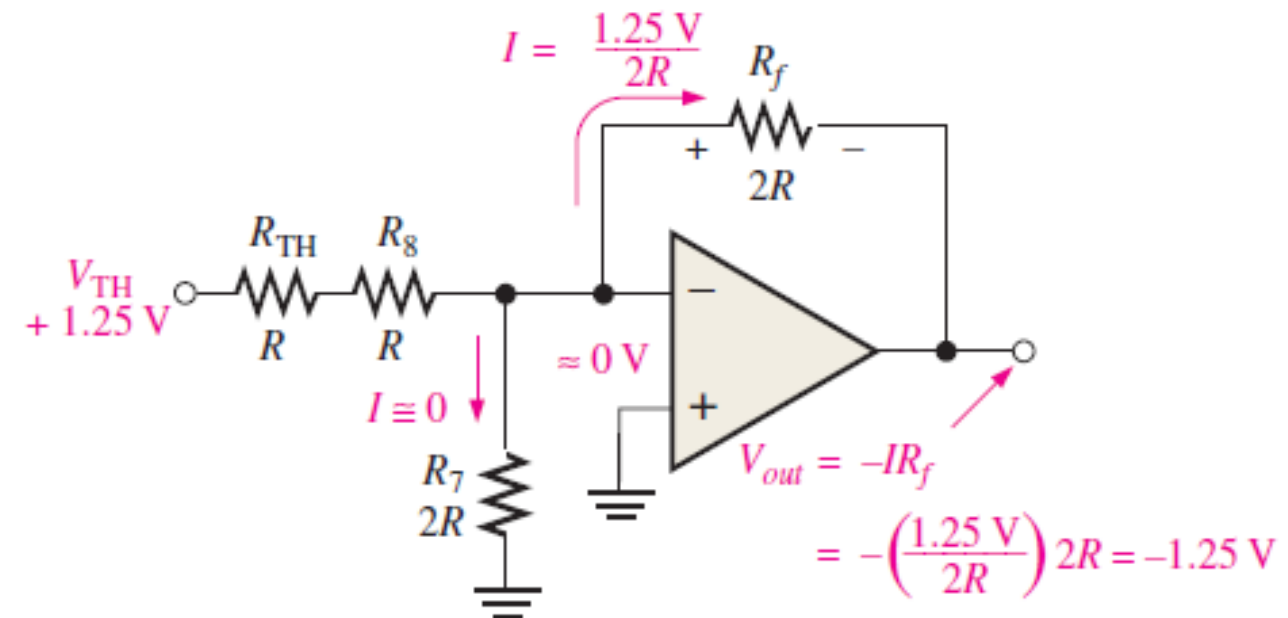


R-2R Ladder DAC

Contribution of D_1 .

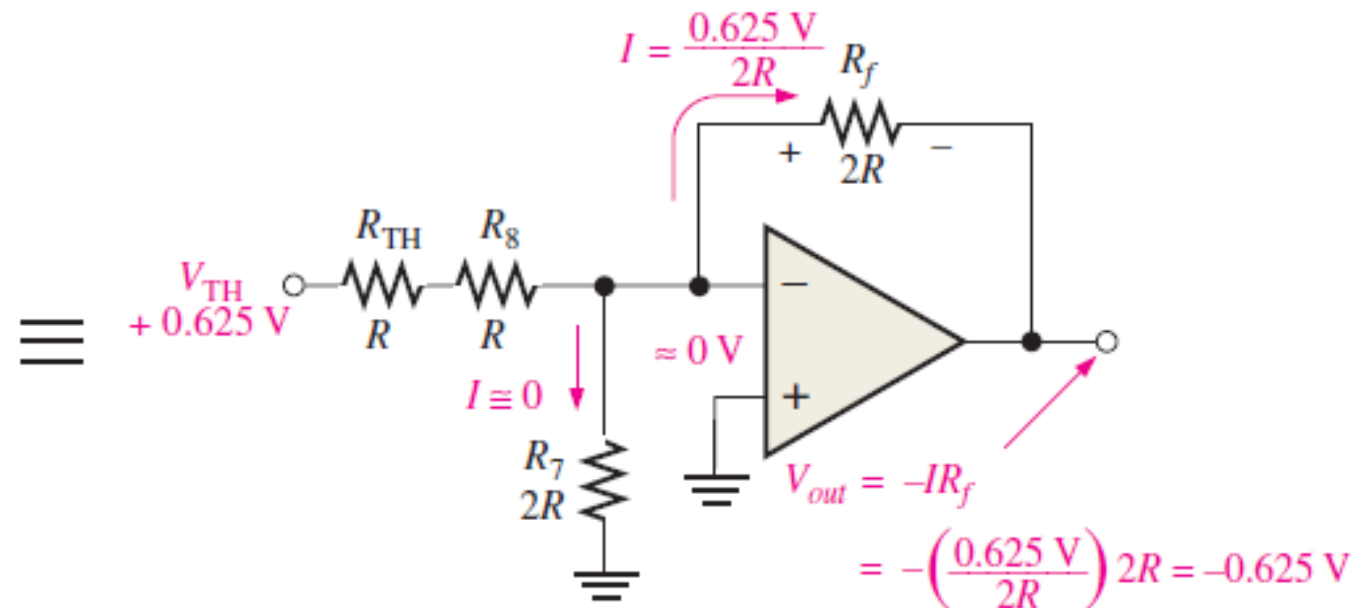
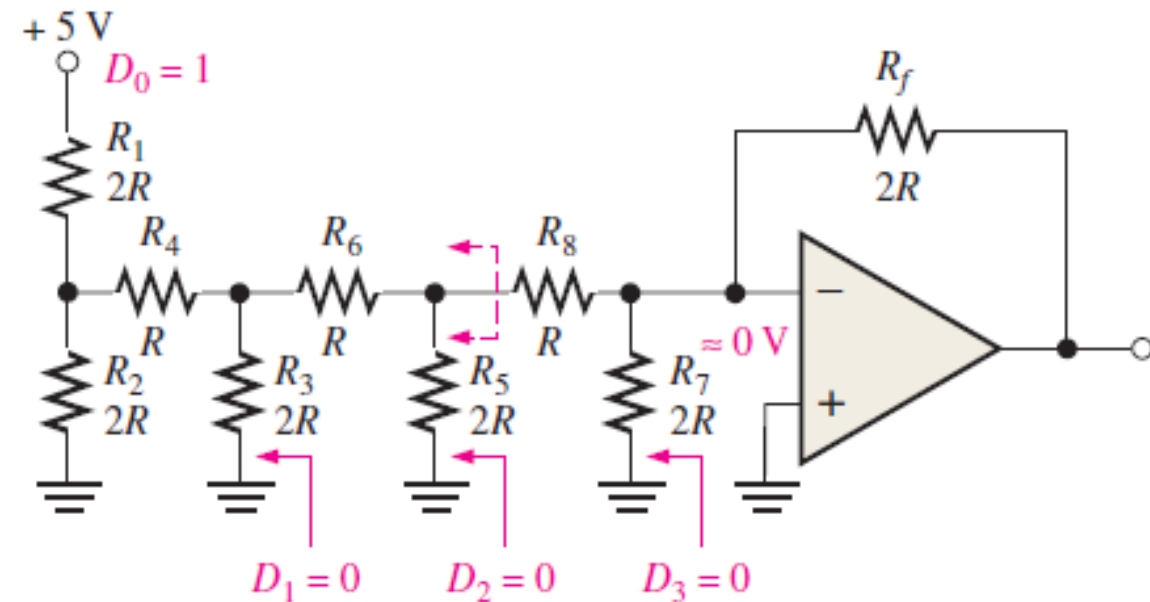
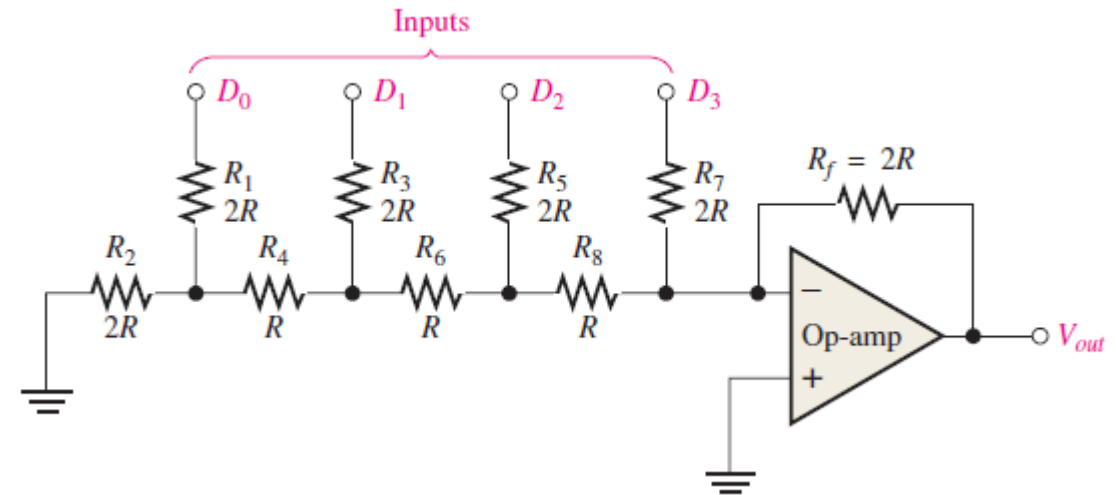


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R-2R Ladder DAC

Contribution of D_0



DAC

Resolution of a n-bit DAC:

$$\frac{1}{2^n - 1}$$

Determine the resolution expressed as a percentage of the following:

- a) An 8-bit DAC
- b) A 12-bit DAC

Solution:

- a) Resolution for 8-bit converter,

$$\frac{1}{2^8 - 1} \times 100 = \frac{1}{255} \times 100 = 0.392\%$$

- b) Resolution for 12-bit converter,

$$\frac{1}{2^{12} - 1} \times 100 = \frac{1}{4095} \times 100 = 0.0244\%$$

1. Thomas L. Floyd, “Digital Fundamentals” 11th edition, Prentice Hall – Pearson Education.

Thank You